

COMMUNITY PERSPECTIVE

Skills need **SemVer** too

What happens after agents can discover them?

Pedro Rodrigues · Supabase

Where Agent Skills are today

The format is portable

```
my-skill/  
├─ SKILL.md  
├─ references/  
├─ assets/  
└─ scripts/
```

Distribution is not standard yet

```
Git repository  
Client-specific installer  
Marketplace or directory  
Over MCP
```

OPEN PR #254

Publisher-origin discovery

<https://supabase.com/.well-known/agent-skills/index.json>

```
{
  "skills": [
    {
      "name": "supabase",
      "type": "archive",
      "url": "/skills/supabase.tar.gz",
      "digest": "sha256:abc123..."
    }
  ]
}
```

Publisher → index → artifact → verified installation

What the proposal already solves

```
Installed: sha256:abc123...  
Current:   sha256:def456... ← changed
```

"A larger benefit is being able to check if any skills have been updated with a single GET."

Jonathan Hefner, PR #254

ORIGIN DISCOVERY

ARTIFACT IDENTITY

CHANGE DETECTION

INTEGRITY

ASSUMPTIONS

In this talk, we're assuming

- 1 .well-known, or something close to it, lands.
- 2 Publishers and clients adopt it.

What is SemVer?

MAJOR.MINOR.PATCH

SEGMENT	MEANING
MAJOR	Breaking change
MINOR	Backward-compatible addition
PATCH	Backward-compatible fix

npm and most package registries already use SemVer as the default versioning convention.

Skills are not npm packages

NPM PACKAGE	AGENT SKILL
Executable code	Mostly instructions and knowledge
A fairly explicit API	A behavioral, probabilistic effect
Compatibility you can test structurally	Depends on model, client, tools, and task
Latest can break the API	Latest often carries the best guidance

For skills, following latest is often the right call.

Why versions, if latest is usually best?

Historical identity earns its keep for:

- A/B testing
- Regression reproduction
- Auditing agent inputs
- Rollback after unexpected behavior

OPTION 1

Digest + URL + lockfile

```
skill: supabase  
url:   .../supabase.tar.gz  
digest: sha256:abc123...
```

Version pinning is done by digest and artifact URL. Essentially, the lock file contains the snapshot of index.json.

The fresh-client problem

Existing client

Observed release → its lockfile can remember it.

Fresh client

Never-seen release → how does it discover the artifact?

It does nothing for a fresh client that wants v0.1.6 of something it has never seen.

CASE STUDY

"with skill" means "with latest"

Today

Without skill ↔ With latest skill

Potential comparison

Skill v0.1.6 ↔ Skill v0.1.7

Reproducible input: skill + release + digest + model + client + tools

One example of a broader lifecycle problem.

Latest can still be worse for one client

Publisher eval environment $\neq$ every consumer environment

- Different model
- Different agent client
- Different tools and permissions
- Different surrounding instructions
- Probabilistic execution

Representative evals are evidence, not a guarantee.

Three signals, three questions

Version

What change did the publisher signal?

Digest

What exact bytes did we use?

Eval

What behavior did we observe?

None of these three questions has the same answer.

Optional historical resolution

```
/.well-known/agent-skills/index.json  
/.well-known/agent-skills/versions.json  
/.well-known/agent-skills/v0.1.6/index.json  
/.well-known/agent-skills/v0.1.7/index.json
```

```
skills install supabase@0.1.6  
skills install supabase@0.1.7
```

Optional for providers · one resolution model · any artifact host

Latest by default. Reproducible when needed.

Merge discovery independently. Keep latest easy. Make historical comparison possible.

Version, digest, and eval each answer a different question.

Open Questions

- Is a lockfile sufficient?
- Who retains history: client, publisher, or registry?
- Do versions belong in SKILL.md, distribution metadata, or both?
- Is SemVer the right language for behavioral change?
- How long should old artifacts stay available?

THANK YOU

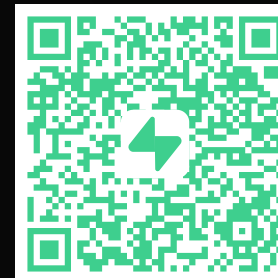
Let's keep the discussion going

Find PR #254 on the Agent Skills repository.

Pedro Rodrigues · Supabase

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SCAN FOR PR #254